

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2025

Course: B. Tech.

Branch: Civil Engineering

Semester: VIII

Subject Code & Name: BTCESS802D Mechanical Characterization of Bituminous Materials

Max Marks: 60

Date:13/ 01/ 2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	Rutting in bituminous pavements is primarily associated with	CO1	1				
	<table border="1"> <tr> <td>A) Low temperature cracking</td> <td>B) Permanent deformation under repeated loads</td> <td>C) Binder oxidation</td> <td>D) Surface polishing</td> </tr> </table>	A) Low temperature cracking	B) Permanent deformation under repeated loads	C) Binder oxidation	D) Surface polishing		
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2	Elastic response of a material is characterized by	CO1	1				
	<table border="1"> <tr> <td>A) Time dependent strain</td> <td>B) Instantaneous and recoverable strain</td> <td>C) Permanent deformation</td> <td>D) Flow behavior</td> </tr> </table>	A) Time dependent strain	B) Instantaneous and recoverable strain	C) Permanent deformation	D) Flow behavior		
A) Time dependent strain	B) Instantaneous and recoverable strain	C) Permanent deformation	D) Flow behavior				
3	Fatigue cracking in bituminous layers mainly occurs due to	CO1	1				
	<table border="1"> <tr> <td>A) Repeated tensile strain at bottom of layer</td> <td>B) High binder content</td> <td>C) Moisture damage</td> <td>D) Excessive air voids</td> </tr> </table>	A) Repeated tensile strain at bottom of layer	B) High binder content	C) Moisture damage	D) Excessive air voids		
A) Repeated tensile strain at bottom of layer	B) High binder content	C) Moisture damage	D) Excessive air voids				
4	Complex modulus in frequency domain is related to	CO2	1				
	<table border="1"> <tr> <td>A) Static loading</td> <td>B) Dynamic loading</td> <td>C) Thermal cracking</td> <td>D) Moisture damage</td> </tr> </table>	A) Static loading	B) Dynamic loading	C) Thermal cracking	D) Moisture damage		
A) Static loading	B) Dynamic loading	C) Thermal cracking	D) Moisture damage				
	Thermorheologically simple materials follow						

5	A) Linear elastic law	B) Time–temperature superposition	C) Plastic flow rule	D) Arrhenius law only	CO2	1
6	Bitumen obtained from refinery behaves mainly as				CO2	1
	A) Pure elastic solid	B) Newtonian fluid	C) Viscoelastic material	D) Brittle solid		
7	Penetration test is conducted at				CO3	1
	A) 35°C	B) 60°C	C) 0°C	D) 25°C		
8	Polymer modification of bitumen improves mainly				CO3	1
	A) Workability only	B) Elastic recovery and rutting resistance	C) Density	D) Aggregate coating		
9	Mixing temperature depends mainly on				CO4	1
	A) Aggregate size	B) Binder viscosity	C) Traffic volume	D) Layer thickness		
10	Van der Poel stiffness modulus considers				CO4	1
	A) Dynamic loading only	B) Time and temperature effects	C) Fatigue cracking	D) Moisture damage		
11	Flow number test evaluates				CO5	1
	A) Fatigue cracking	B) Moisture damage	C) Thermal cracking	D) Rutting resistance		
12	Repeated indirect tensile test induces				CO5	1
	A) Pure shear stress	B) Only compressive stress	C) Thermal stress	D) Tensile stress		
Q. 2	Solve the following.					12
A)	Explain the basic structure and components of a bituminous pavement.				CO1	6
B)	Explain distresses such as rutting and fatigue cracking in bituminous layers.				CO1	6

Q.3	Solve the following.		12
A)	State the Boltzmann's Superposition Principle and its use in pavement engineering	CO2	6
B)	Explain the complex creep compliance and complex stress relaxation functions.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Describe aging of bitumen and bituminous mixtures.	CO3	6
B)	Explain the difference between Penetration grading and Viscosity grading systems.	CO3	6
C)	Write a description about chemical composition of bitumen.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Write a short note on the Dynamic Shear Rheometer (DSR) and its uses.	CO4	6
B)	Discuss issues related to reliability, slow-moving traffic, and grade-bumping.	CO4	6
C)	Introduce the Multiple Stress Creep and Recovery (MSCR) test method.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Describe the volumetric properties important for bituminous mixtures.	CO5	6
B)	Explain in detail causes of failure of bitumen pavement.	CO5	6
C)	Explain mixing and compaction of unmodified and modified bituminous mixtures.	CO5	6
	*** End ***		